

18 Carboxylic acids and derivatives

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Carboxylic acids donate H^+ because carboxylate is resonance stabilised. Esters form by acid-catalysed condensation of acid and alcohol; hydrolysis reverses ester formation. Acyl chlorides are more reactive than esters and undergo nucleophilic acyl substitution, giving HCl with alcohols or water.

Worked example: follow the reasoning

Ethanoic acid + ethanol $\rightleftharpoons$ ethyl ethanoate + water under concentrated H_2SO_4 catalysis.

Quick check · multiple choice

A What is produced when ethanol reacts with ethanoic acid under suitable conditions? [1]

- A Ethanamide
- B Ethyl ethanoate
- C Ethanoyl chloride
- D Ethene

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Name ester from propanoic acid and methanol, and state two conditions/features of preparation. [3]

Answers and explanation

Quick check: B. Ethyl ethanoate

Question 1 [3 marks]

Methyl propanoate [1]; acid catalyst e.g. concentrated H_2SO_4 [1]; heat under reflux/reversible condensation with water [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

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